

A Post-Processing Adapter for Correcting False Positives and False Negatives in Egocentric Hand Detection

Detection Quality Adapter Project
Egocentric hand-detection post-processing pipeline

Abstract—Object detectors operating on video make two kinds of mistakes that are invisible in a single frame: reporting a hand that is not present (a false positive) and missing a hand that is (a false negative). Both are detectable only across multiple frames, since a real hand follows a continuous, physically plausible trajectory that noise does not. This paper describes a detection quality adapter: a stateful, six-stage post-processing layer inserted between a hand detector and its downstream consumers, which corrects both error types using per-frame geometry and multi-frame tracking, without retraining or otherwise modifying the detector. The adapter was built and calibrated against 39 real egocentric work clips (85,784 frames, 153,876 raw detections) spanning 39 distinct jobs, for which no labeled ground truth exists. In the absence of labels, every threshold was checked against the dataset’s own real distributions rather than set by inspection, and the pipeline’s behavior is further validated through three label-free evaluations: masked-detection recovery, track-structure analysis, and per-rule ablation with threshold sensitivity sweeps. The final calibrated pipeline retains 85.3% of detections, recovers 2.5% through interpolation, and rejects the remainder for cause; masked short-to-medium dropouts (1–10 frames) are recovered at 76–78% with gracefully degrading localization accuracy. A depth-based rule for separating a wearer’s hands from a bystander’s was found not to generalize across recording batches and is documented as opt-in rather than shipped by default.

Index Terms—hand detection, multi-object tracking, false positive rejection, temporal consistency, egocentric video, stereo depth, post-processing pipeline, label-free evaluation, ablation study

I. Introduction

A hand detector scoring video frame by frame fails in two directions. A false positive is a detection reported where no hand is present: a duplicate box, a bystander’s hand, a shadow mistaken for skin. A false negative is a hand present in the frame that the detector fails to report, typically during motion blur or a momentary bad frame. False positives retain footage that should be discarded; false negatives discard footage that was sound.

Neither error is visible from a single frame. A duplicate box looks like a normal detection in isolation. A missed hand looks exactly like an empty frame. The only signal that separates a real detection from a spurious one, or a genuine gap from a real absence, is behavior across many frames: a real hand moves along a continuous, physically plausible path, and noise does not.

The straightforward response — retrain the detector — does not fix the structural problem. A detector scoring one frame at a time has no way to know that a duplicate should lose to the real hand, because within that single frame it cannot tell them apart. This work takes the opposite approach: let the detector over-report (every plausible box, uncapped, unduplicated), and place a stateful, multi-frame adapter after it that has context a single frame never will. This reframes the problem as one of tracking and consistency, not computer vision. Nothing in the adapter performs object detection; every stage asks, given noisy boxes over time, what is the most plausible explanation.

The adapter is specified in two parts. Part 1 defines a generic, class-agnostic version: four stages of geometric and temporal correction that require nothing more than per-class size, shape, and speed expectations. Part 2 specializes it for hands with a low, constantly-reached instance cap, a head-mounted camera’s near-continuous motion, a characteristic downward exit path, and a second camera view that makes per-box depth available. This paper covers both parts as built, the real dataset used to calibrate them, what could and could not be validated in the absence of labeled ground truth, and three label-free evaluations designed to characterize the pipeline’s behavior without requiring annotation.

II. Problem Formulation

The adapter sits between a detector and everything downstream, and its contract is deliberately narrow, as summarized in Table I.

Nothing is ever deleted. A rejected box stays in the record, tagged untrustworthy rather than removed, for two reasons: every stage’s decision remains auditable after the fact, and a later stage can revise an earlier tag — interpolation, for instance, can overwrite a rejected box’s position when the surrounding trajectory supports it. An interpolated detection is an assertion the adapter made that the detector never did, and downstream consumers must be able to tell the two apart.

Six classes of false positive are separable without a model, split by whether they need tracking, listed in Table II.

The split fixes the pipeline’s order: the last three rules need tracks, and tracks only exist after association, so

TABLE I
The Adapter’s Contract, as Specified

Aspect	Specification
Consumes	Per frame, a candidate pool of boxes with confidences, plus frame timestamps
Returns	Corrected detections, each marked reported, merged, rejected, or interpolated
Configured by	Per-class expectations: plausible size, plausible shape, candidate pool size and class maximum, maximum speed, and typical exit behavior
Does not	Alter the detector or require retraining. It does ask the detector to stop capping instances itself, and takes over that selection

TABLE II
The Six Separable False-Positive Classes and Their Order of Application

#	Class	Basis	Rejection Criterion
1	Duplicate detections	Geometric	Two heavily overlapping boxes on one object — merge, retain the stronger
2	Implausible size	Geometric	Far larger or smaller than the object can be at working distance
3	Implausible shape	Geometric	Markedly more elongated than the object should be
4	Implausible displacement	Temporal	The box jumps further between consecutive frames than the object can travel
5	Unsupported detection	Temporal	Appears for one or two frames with no track before or after it
6	Static detection	Temporal	Does not move while the camera does, indicating scene structure

geometric rejection must run before association, which must run before temporal rejection, which must run before interpolation. This ordering is also why the instance cap is deferred rather than applied at the detector. A detector asked to cap itself selects by confidence within a single frame — exactly the comparison that is unreliable, since a duplicate or background object can outscore a real detection in a given frame, discarding the real one before the adapter ever sees it. Instead the detector emits a slightly wider candidate pool, and the adapter selects after association, ranking candidates by the track supporting them rather than by single-frame score. This admits more false positives at the input, which is acceptable because the temporal rules exist to remove them — but the pool stays narrow, since a wide one only gives association more spurious candidates to confuse.

III. System Architecture

Detections flow through six stages, always in this order (below). Stages 5 and 6 are hand-specific additions layered on top of the four-stage generic core.

1 GEOMETRIC REJECTION — Duplicate merge → size filter → shape filter. Per-frame only.
2 ASSOCIATION (TRACKER) — Greedy nearest-neighbor match against each track’s velocity-predicted position.
3 TEMPORAL REJECTION — Implausible displacement, unsupported flicker, static-against-moving-camera.
4 INTERPOLATION AND EXIT DETECTION — Exit test first; otherwise fill the gap if resumption matches prediction.
5 INSTANCE-CAP SELECTION (hand-specific) — Keep the two best-supported tracks, not the two highest-confidence boxes.
6 STEREO DEPTH (hand-specific, opt-in) — Reject detections farther than arm’s reach. Excluded from the default output.

Fig. 1. The six-stage adapter pipeline, applied top to bottom. Order is load-bearing: stages 3–6 all require tracks, which do not exist until stage 2 has run.

Every detection carries a tag — reported, merged, rejected, or interpolated — that survives to the final output, making every stage’s decision inspectable after the fact.

IV. Key Design Decisions

A. Tracking Algorithm: Greedy Nearest-Neighbor, Not Kalman or Hungarian

The tracker predicts each track’s next position by linear extrapolation from its last known velocity, then matches the nearest detection within a distance gate. This is deliberately the simplest approach that could work; a Kalman filter (for a smoothed velocity estimate under sensor noise) and Hungarian matching (for globally optimal assignment instead of greedy) were flagged as upgrade paths in code comments but not built, because the simple version passed every synthetic and real-data test. Premature sophistication is a cost, not a feature — the upgrade is built when a real failure demands it, not speculatively.

B. Identity Signal: Position and Motion Only, Never the Detector’s Left/Right Label

The detector guesses left- or right-hand per box, but that label is least reliable exactly when it matters most: when two hands cross. The tracker ignores it entirely for identity decisions, verified with an adversarial test in which two hands cross with their labels swapped at the crossing point. An unreliable signal is worse than no signal at all if the system does not know it is unreliable.

C. Two Gate Radii, Not One

Early versions used one shared `max_speed` value to govern both the tracker’s match gate and Stage 3’s displacement-plausibility check. Real data showed this to be wrong: a genuinely fast hand movement needs a generous gate so the tracker does not fragment the track, but that same movement should still be flagged as visually untrustworthy once it is part of a track. Conflating “should these be considered the same object” with “is this specific frame trustworthy” is a common tracking bug — they are different questions on different timescales. The fields were split into `track_gate_speed_px_per_frame` (350, generous) and `max_speed_px_per_frame` (110, tightened); see Section VII for the effect this had on real tracks.

TABLE III
Dataset Composition

Quantity	Value
Clips	39
Total frames	85,784
Total raw detections	153,876
Ground-truth labels available	None
2 boxes/frame	72.3%
1 box/frame	18.2%
3+ boxes/frame	5.25%
0 boxes/frame	4.3%

D. A Generic Core Plus a Thin, Config-Only Hand Layer

Stages 1–4 know nothing about hands specifically; they are written for any tracked object with configurable size, shape, and speed expectations. A `hand_config()` function supplies hand-specific numbers, and two additive stages (selection, stereo depth) sit on top, without any change to stages 1–4. This is what keeps the generic core testable in isolation and reusable for a different object class.

E. Calibration Philosophy: Real-Data Checks Over Guesses, Labels Over Both

Every placeholder threshold was checked against the real dataset’s own distributions before being trusted, using percentiles and structural invariants rather than visual inspection. A threshold that was a complete no-op or actively wrong was fixed immediately, without waiting for labeled ground truth. Anything that genuinely needed labels — true precision and recall, candidate-pool-width tuning — was left alone and documented as blocked, rather than guessed at. The absence of labels is not the same as an absence of anything checkable, a claim this paper substantiates further in Section VIII with three evaluations that require no annotation at all.

V. Dataset

39 clips, each a distinct real work task, recorded on a head-mounted stereo camera at 1920×1200, 30 fps, both eyes. Every clip bundle carries five artifacts: the stereo video pair, raw hand boxes from a fine-tuned detector (uncapped, unduplicated, unsmoothed — deliberately noisy), full-rate 6-DoF camera pose from onboard visual-inertial odometry, per-frame timestamps, and task meta-data.

Remark: every one of the 39 clips is a different job — hairstylist, carpenter, farmworker, jeweler, mason, mechanic, and others, no repeats (independently verified: 39 unique values in meta.json’s job field across all 39 clips). This is why several “one global number” thresholds turned out to be too simple: a farmworker’s typical working distance (median 0.77 m) and a hairstylist’s (1.5–1.8 m, arms extended toward a seated client) are genuinely different populations, not noise around one true value. This recurs throughout Sections VI–VII and again in Section VIII.

TABLE IV
Hand-Specific Properties and Their Consequences

Property	Consequence for the Adapter
At most two instances per frame	The cap is low and constantly reached; the detector requests 3–4 candidates, the adapter selects two after association.
Head-mounted camera	The camera moves constantly, so the static-detection rule is unusually discriminating — with an important exception (Section VI-B).
Characteristic exit path	Hands most often leave downward; the leaving-the-frame test weights the lower border accordingly.
A second eye is available	Per-box stereo depth separates the wearer’s hands from a bystander’s — the strongest discriminator available, with no equivalent in the generic adapter.

VI. Hand Specialization

Hands are one configuration of the generic adapter. Most of the specialization is parameters rather than new logic, but four properties change behavior materially (Table IV).

Two further properties are easy to assume incorrectly. Hands are roughly equant, so the shape rule is informative here where it would not be for an elongated object. And handedness is unreliable exactly when hands cross, so tracking associates on position and motion, never on the reported label (Section IV-B).

A. Duplicate Detection vs. Crossing Hands

The generic dedup rule (Stage 1) merges on IoU alone, before any tracking exists — it has no way to know whether two overlapping boxes are one object detected twice, or two real hands that happen to overlap. A real-data check across all 39 clips found the discriminator: every real duplicate pair had a containment ratio (intersection over the smaller box’s area) of at least 0.73 — one strong box mostly nested inside a weaker one — while a synthetic same-size 50%-overlap crossing sits at 0.5. A `duplicate_containment_threshold` of 0.7 was added for hands, verified against a pair chosen deliberately in the gap between the two gates (IoU 0.52, containment 0.68).

B. Where the Static-Detection Assumption Breaks

The specification’s assumption is that a head-mounted camera moves constantly enough that a hand is never stationary in the image for a sustained period. One carpentry clip contradicts this directly: a hand braced motionless for several seconds while steadying a work-piece, despite continuous camera motion. Requiring 15 consecutive still-while-moving frames before rejecting a detection (rather than one) resolved brief real pauses, but not this multi-second brace. The underlying limitation is structural: telling a genuine multi-second pause from true background clutter needs a model that separates rotation-induced apparent motion from real parallax, not

TABLE V

Thresholds Retuned from Real, Unlabeled Data Across All 39 Clips

Parameter	Before	After	Evidence
plausible_size	(20,800)px	(50,1150)px	p1=94px, max=1110px
plausible_shape	(0.3,3.0)	(0.5,2.0)	p1=0.54, p99=2.10
max_dropout_frames	15	15	gap-length p75=13 frames
max_speed (plausibility)	150 px/f	110 px/f	speed p99.5=97.7
track_gate_speedshared		350 px/f	above fastest jump (305.8)
duplicate_containment	0.7	0.7	duplicates ≥ 0.73 , crossings ~ 0.5

a longer or shorter pixel-threshold window. Section VIII-C quantifies this rule’s overall impact directly: it is the single largest contributor to rejections of any rule in the pipeline.

C. Stereo Depth: Validated on One Clip, Not on the Dataset

The “beyond arm’s reach” rule (Stage 6) was calibrated from a nominal ZED X stereo model — 2.2 mm lens, 12 cm baseline, 733 px focal length — cross-checked against a real high-confidence box at frame 230 of one clip, where a disparity of roughly 267 px implied a depth of about 0.33 m, consistent with that lens’s minimum focus distance for a box that large. Running the same calibration across all 39 clips surfaced a confident false rejection of the wearer’s own hands on a different clip, traced to the dataset spanning multiple recording batches with different metadata signatures. Stage 6 is therefore opt-in, invoked only when both video paths are explicitly supplied, and excluded from the pipeline’s default and final output.

VII. Calibration and Results

The specification’s validation section asks for two things: precision and recall computed per stage, and a labels-free standing check on the proportion of interpolated detections. Precision and recall are both defined against ground truth — whether a box was actually a hand — and this dataset has none. The matching machinery to compute them is fully built and unit-tested against synthetic ground truth, ready the moment a labeled reference set exists, but it produces no real figure today. What follows is what was honestly achievable without labels; Section VIII then goes further, using three evaluations designed specifically to probe pipeline behavior without requiring any ground truth at all.

A. Threshold Tuning From Real Distributions

Six thresholds were retuned using percentiles of the real dataset rather than inspection (Table V).

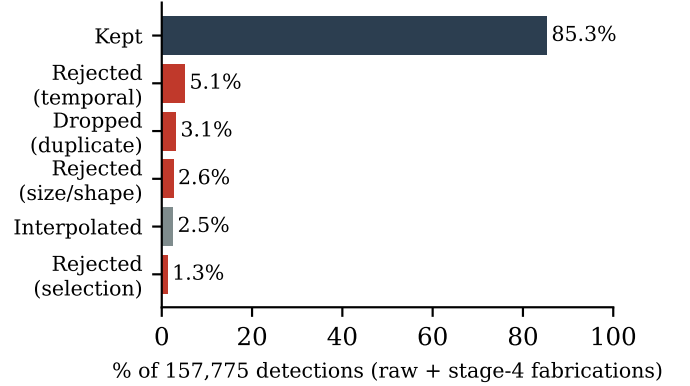


Fig. 1. Detection outcome breakdown across all 39 clips, reproducible via `python calibration/sweep_thresholds.py --data-dir data` (runtime: ~ 4 seconds).

TABLE VI
Final Results Across All 39 Clips

Metric	Value
Final detections (Stages 1–5)	148,714
Kept	90.5%
Rejected — temporal (Stage 3)	5.5%
Interpolated (Stage 4)	2.6%
Rejected — selection (Stage 5)	1.4%
Dropped as duplicate (Stage 1)*	3.1%
Rejected — size/shape (Stage 1)*	2.6%
Tracks confirmed exiting	36.1%
Interpolated proportion, mean/max	3.5% / 17.2%

*Measured against all 153,876 raw detections.

B. Where Every Detection Ends Up

Across all 39 clips, 153,876 raw detections plus 3,899 Stage-4 fabrications (157,775 total) give the breakdown in Fig. 1 by rejection reason — a real, unsupervised proxy for where each rule fires, not a confirmed false-positive rate.

C. Final Calibrated Pipeline

The tuned hand pipeline (Stages 1–5, Stage 6 excluded) produces 148,714 final detections across 787 tracks (Table VI).

VIII. Label-Free Behavioral Evaluation

Rejection-reason frequencies (Section VII) describe where the pipeline acts but not whether it acts correctly. This section adds three evaluations that probe pipeline behavior using signals already present in the data — without requiring any manual annotation. Each is explicit about what it does and does not establish; none of them substitutes for true precision/recall against ground truth.

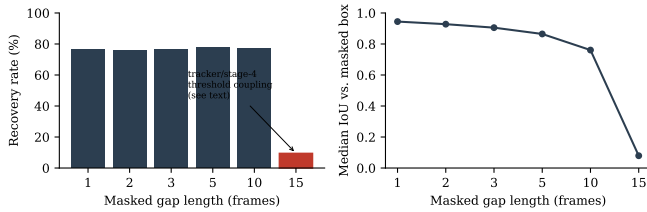


Fig. 2. Recovery rate and median IoU vs. masked gap length, pooled across all 39 clips (1,061–1,815 placements per gap length). Reproducible via `python calibration/dropout_recovery.py --data-dir data`.

A. Dropout Recovery

Method. For each clip, we identify runs of consecutive, unmodified detections within a track and artificially delete interior blocks of length 1, 2, 3, 5, 10, and 15 frames, using one centered placement per qualifying run. The pipeline (Stages 1–4) is then re-run on the masked input from a fresh clip load, and we measure: the recovery rate (does an interpolated detection reappear at each masked frame), localization accuracy (IoU and center distance against the deleted box), and the false-fill rate (spurious interpolation at unmasked neighboring frames). The deleted box itself is the detector’s own (already stage-1–3-filtered) output, not ground truth, so this measures self-consistency of the recovery mechanism, not correctness against real hand position.

Results. Recovery rate is flat at 76–78% for gaps of 1–10 frames, while localization quality degrades smoothly as the gap widens (median IoU 0.945 at gap 1 down to 0.761 at gap 10; Fig. 2). At gap 15, recovery collapses to 9.5%. This is not noise: at a 15-frame masked gap, the true distance between surviving anchors is 16 frames, which exceeds `max_dropout_frames` (15) — a config field shared between the tracker’s own patience window (association.py) and Stage 4’s fill cutoff (interpolation.py). The tracker itself expires the track before Stage 4 is ever consulted; the collapse is a config-coupling artifact, not evidence that 15-frame interpolation is inherently unrecoverable. False-fill rate stays under 1.2% throughout.

What this shows. The interpolation mechanism recovers a large, consistent majority of short-to-medium dropouts with position/size accuracy that degrades gracefully rather than catastrophically, and does so with a low fabrication rate. It also surfaces a genuine, previously undocumented coupling: sharing one threshold between the tracker’s patience window and the interpolation cutoff creates a hard reliability cliff exactly at that threshold. What it does not show: correctness against real hand position — a detector systematically wrong in the same way at a masked frame and at its own interpolation of that frame would score perfectly here.

B. Track Structure

Method. We compare “association only” (Stages 1–2) against the full hand pipeline (Stages 1–5) across all 39 clips, measuring track count, mean track length, a fragmentation proxy (a track starting within 30 frames and $2\times$ box diagonal of another track’s end — a plausible, not confirmed, severed continuation), gaps per track, and interpolated proportion.

Results. Track count (787), mean track length, and the fragmentation proxy (254 candidates, 28.5% of tracks) are identical between the two arms. This follows directly from the pipeline’s design: no stage after the tracker (Stage 2) ever merges, splits, or extends a track’s start/end — Stages 3–5 only tag existing detections or fill gaps strictly between them. What the correction stages change is track content: total gap-frames drop by 53% ($8,240 \rightarrow 4,341$) and interpolated proportion rises from 0.0% to 3.5%.

What this shows. The temporal/interpolation/selection stages operate purely as content-level correctors on top of whatever tracks association already produced — they cannot be masking poor tracking by artificially bridging fragmented tracks into fewer, longer ones, because track topology is provably unchanged. What it does not show: whether the association stage’s own 28.5% fragmentation rate is itself good or bad — the proxy is a spatial/temporal heuristic, and two distinct real objects passing through the same location in quick succession would trigger it identically to a genuinely severed track.

C. Ablation and Sensitivity

Method. Each of the eight per-frame/per-track rules is disabled in turn via a structural no-op configuration override (e.g. `duplicate_iou_threshold=1.1`, an unreachable IoU), and the full 39-clip pipeline is re-run from a fresh load for each configuration (runtime: 10.7 s for all eight ablations). Separately, eleven thresholds are swept across 8–9 points spanning a plausible range centered on their current default (85 full-dataset passes, 116.7 s total), reporting how kept% responds.

Ablation results. Fig. 3 ranks the eight rules by their effect on kept% when disabled. The static-detection rule (Section VI-B) is the single largest contributor (+4.8 pp when disabled), followed by the shape filter (+2.5 pp); the size filter and flicker/unsupported checks are structurally near-inert (0.0 pp) on this dataset. Disabling dedup reduces kept% (−2.0 pp) despite eliminating duplicate rejections outright, because unduplicated boxes then compete for Stage 5’s fixed 2-instance cap and lose there instead — evidence that dedup and selection perform complementary, not redundant, work. `max_dropout_frames` could not be cleanly ablated for Stage 4 in isolation, since the tracker reads the same field for its own patience window (Section VIII-A); disabling it (setting it to 0) collapses the tracker itself rather than isolating interpolation, and is excluded from Fig. 3 for that reason.

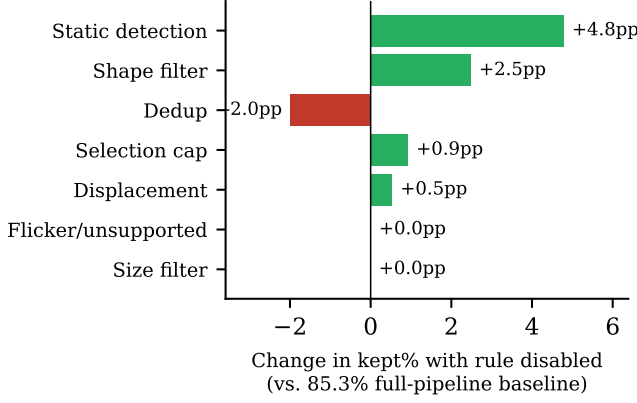


Fig. 3. Change in kept% when each rule is disabled, vs. the 85.3% full-pipeline baseline. Reproducible via python calibration/ablation.py -data-dir data.

TABLE VII
Threshold Sensitivity Verdicts

Threshold	Default	Verdict
max_speed_px_per_frame	110	flat
track_gate_speed_px_per_frame	350	flat
max_dropout_frames	15	knife-edge at 0→1 only
min_static_run_frames	15	knife-edge
static_px_threshold	4.0	knife-edge
duplicate_iou_threshold	0.5	flat
duplicate_containment_threshold	0.7	flat
plausible_size (lower)	50	knife-edge above ~120px
plausible_size (upper)	1150	flat
plausible_shape (lower)	0.5	knife-edge above ~0.5
plausible_shape (upper)	2.0	knife-edge below ~2.0

Sensitivity results. Table VII and Fig. 4 classify eleven thresholds as sitting on a flat plateau (small kept% change across the swept range) or a knife-edge (a single-step change exceeding 5 pp anywhere in range). `track_gate_speed_px_per_frame`, `duplicate_iou_threshold`, `duplicate_containment_threshold`, and `plausible_size`’s upper bound are flat — their current defaults could move substantially with negligible effect. `static_px_threshold`, `min_static_run_frames`, `plausible_size`’s lower bound, and both `plausible_shape` bounds are knife-edge: exactly the thresholds where a labeled reference set would matter most, since small changes there swing outcomes by double-digit percentage points.

What this shows. Without labels, ablation and sensitivity together identify which rules and thresholds materially drive outcomes on this dataset versus which are structurally inert or robust to their exact setting — a defensible, if informal, robustness claim for the flat thresholds, and

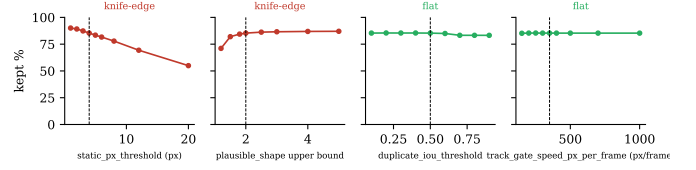


Fig. 4. Representative sensitivity curves: two knife-edge thresholds and two flat thresholds. Dashed line marks the current default. Reproducible via python calibration/ablation.py -data-dir data.

TABLE VIII
Summary of Limitations

Limitation	Detail
No ground truth	Every number in this paper, including the label-free evaluations of Section VIII, is evidence-based or self-consistency-based, not precision/recall-optimal.
Stereo depth does not generalize	Calibrated and validated on one clip; misjudged depth by roughly 4× on a different clip. Opt-in, excluded by default.
Multi-second static braces	The largest single driver of rejections (Section VIII-C) cannot yet distinguish a multi-second deliberate hold from background structure.
max_dropout_frames, cross-stage coupling	The same field gates both the tracker’s patience and Stage 4’s fill cutoff, producing a sharp recoverability cliff at the threshold rather than graceful degradation (Section VIII-A).
One global number for 39 jobs	Arm’s reach, camera speed, and hand size vary by task; a job-aware version was considered and not built, since deriving a job’s normal range from its own detections would be circular without labels.
Greedy, not globally optimal tracking	Nearest-neighbor matching can in principle mis-assign in dense, ambiguous scenes; not observed in this dataset, but a known theoretical gap.

a concrete prioritization signal for where future labeled calibration effort matters most. What this does not show: correctness. A rule firing more or less does not by itself indicate it is now more or less accurate; “knife-edge” flags where the optimal value would matter, not what that value is.

IX. Limitations

Table VIII summarizes known limitations, updated with the findings from Section VIII.

X. Conclusion and Future Work

This paper described a six-stage post-processing adapter that corrects false positives and false negatives in hand detection using temporal consistency rather than detector retraining. Beyond the rejection-reason frequencies used to characterize where the pipeline acts, three label-free

evaluations characterize how well it acts without requiring any manual annotation: dropout recovery shows a consistent 76–78% recovery rate with gracefully degrading localization for short-to-medium gaps; track-structure analysis shows the correction stages cannot be masking poor tracking, since track topology is fixed entirely by the association stage; and ablation/sensitivity analysis identifies which rules and thresholds materially drive outcomes, prioritizing exactly where future labeled calibration would matter most.

Three lines of work remain, all requiring the labeled reference set the specification calls for: true per-stage precision and recall, candidate-pool-width tuning, and validation of a job-aware scaling factor for the knife-edge thresholds identified in Section VIII-C. A fourth line does not require labels but does require new modeling: separating camera-rotation-induced apparent motion from true parallax, to resolve the multi-second static-brace failure. A fifth — decoupling `max_dropout_frames`’s two roles so that interpolation reliability degrades smoothly rather than cliff-like at the threshold — follows directly from Section VIII-A and requires no labels at all. A sixth — recovering or deriving true per-clip stereo calibration — would let Stage 6 move from opt-in to a validated part of the default pipeline.

Appendix

Glossary of Terms

Term	Meaning
Detection	One raw box the detector reported on one frame, with a confidence score
Track	An ordered sequence of detections believed to be the same physical object over time
Tag	What happened to a detection: reported, merged, rejected, or interpolated
IoU	Intersection-over-union — overlap between two boxes, 0 to 1
Disparity	Pixel shift of the same point between stereo views
VIO pose	Visual-inertial odometry — the camera’s estimated 6-DoF pose per frame
Gate/gating	A maximum-distance cutoff for a plausible track match
Standing check	A labels-free monitor — here, interpolated proportion
Knife-edge	A threshold where a small change produces a large (>5pp) outcome swing
False-fill rate	Rate of spurious interpolation at frames that were not masked/missing